Sahaj Bhadja

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Summary

Embedded Software Engineer with 2+ years building **C++/Python systems**, **integration testing**, concurrent workflows, and **system validation** across Linux, networking, debugging, and automated release pipelines.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **system validation**, reconciliation, error handling, cloud storage integration, CI/CD releases, and stable production monitoring workflows reliably at enterprise scale.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable processing, and observable workflows under sustained production load reliably.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured validation workflows**, model switching, fallback paths, persistent results, and reviewer controls supporting repeatable analytical processing for technical users.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through **system-level simulation** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across diverse autonomous-driving scenarios reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **real-time integration platform** covering authentication, resources, customers, billing, permissions, and notifications with React, Node.js, WebSockets, and Azure SQL across distributed locations.
- Reduced resource wastage by 42% through **interoperability testing**, implementing JWT access controls, WebSocket updates, automated tests, monitoring, and Docker/Jenkins deployments across eight modular product service domains reliably.

Skills

Languages: C++, Python, Bash, Java, SQL, JavaScript
Systems & Networking: Linux, Windows, macOS, Operating Systems, TCP/IP Fundamentals, WebSockets, REST APIs, Distributed Systems, Concurrent Programming, Multithreading, CPU Scheduling
Integration & Validation: Device Integration, Software Verification, System Validation, Interoperability Testing, Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Root-Cause Analysis, Fault Injection, Error Handling, Data Validation
Tools & Delivery: AWS, Azure, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD
Engineering Practices: Object-Oriented Programming, Data Structures, Algorithms, Performance Optimization, Monitoring, Technical Documentation, Agile

Projects

Networked Device Interoperability Harness - Design Concept
- Designed a **TCP/IP device-validation harness** concept in C++ and Python with configurable device profiles, socket simulators, fault injection, packet logs, connectivity diagnostics, automated regression tests, and validation reports.

Firmware Release Validation Console - Design Concept
- Designed a **firmware release-validation console** concept coordinating configuration snapshots, connectivity checks, automated test execution, structured logs, failure triage, corrective-action records, release comparisons, and repeatable technical validation reports.

CPU-Scheduling Simulator | Live
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios.